

ADIP RANJAN DAS

PhD Researcher · Robot Learning, Dual-Arm Manipulation & Perception for Robotic Disassembly

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RESEARCH PROFILE

Robotics PhD researcher at Heriot-Watt University (ARM²Lab, National Robotarium) building autonomous **dual-arm systems for disassembly of end-of-life electronics**. Work spans learned perception, LLM-agent task planning with failure recovery, force-sensed tool manipulation and simulation digital twins, all validated on **physical dual-arm hardware** across three device classes. First-author publications at IEEE/ASME AIM and IEEE Access; first-author IEEE RA-L submission under review. Industry background in robotics product development (defence-funded) and software engineering.

RESEARCH INTERESTS

Robot learning for contact-rich manipulation · reliability, failure detection & corrective action for learned policies · dual-arm coordination · Vision-Language-Action models · perception for disassembly · sim-to-real transfer

EDUCATION

PhD in Robotics — Heriot-Watt University, Edinburgh, UK

Nov 2023 – Present

ARM²Lab, National Robotarium · Supervisors: Dr Maria Koskinopoulou, Dr Xianwen Kong

- **Thesis:** Dual-robot manipulation in industrial settings for disassembly of electronic equipment.
- Addresses under-studied aspects of robotic disassembly in the context of Industry 4.0, developing a fully autonomous dual-arm system that performs disassembly efficiently and adapts across device classes (HDDs, mini PCs, laptops).

MSc in Robotics (Distinction) — University of Bristol, UK

Sep 2022 – Sep 2023

Bristol Robotics Laboratory

- **Dissertation — Occupational Upper-Limb Exoskeleton:** researched exoskeleton design to improve human–robot interaction and workplace safety ([code](#)).
- **Apple Counting with Machine Learning:** applied machine-vision and ML techniques to count apples in orchard imagery, streamlining yield estimation ([report](#)).
- **IR Sensor Communication (team lead):** led a team investigating and optimising IR sensor communication for reliable robot-to-robot following ([report](#)).
- **Line-Following Challenge:** developed and implemented control algorithms for precise line following on an autonomous robot ([video](#)).
- **Socially & Physically Assistive Robots:** in-depth case study of assistive robots, their impact on human–robot interaction and applications across care settings ([report](#)).

BTech in Electronics & Computer Science — KIIT, Bhubaneswar, India

Jul 2016 – Jun 2020

Kalinga Institute of Industrial Technology · GPA 8.5 / 10

- **Major project — Road-Sign Detection for Driving Assistance:** implemented a computer-vision pipeline to detect road signs and traffic signals in support of driver-assistance systems ([report](#)).

RESEARCH EXPERIENCE

Adaptive Dual-Arm Robotic Disassembly of End-of-Life Electronics

Nov 2023 – Present

PhD research, ARM²Lab, Heriot-Watt University · ROS 2 Humble, MoveIt 2, Python, PyTorch

- **Built a dual-arm disassembly cell end to end:** slider-mounted 6-DoF UF850 + OnRobot RG6 (Fin-Ray fingertips) for manipulation; 5-DoF xArm5 with custom screwdriver, Robotiq FT300 force-torque sensor, screw-retaining jaws and endoscopic camera for fastener removal.
- **Designed a three-tier learned perception stack:** RF-DETR-L for global component detection (11 classes), RF-DETR-S for local fastener detection (4 classes) and YOLOv11-n for post-action state verification (3 classes); custom datasets built in Roboflow.
- **Developed an LLM-agent planner** (LangGraph, ReAct loop) over a structured world state of objects, fasteners and events that sequences dual-arm skills (Unscrew, Pick-and-Extract, Flip) and routes detected failures to corrective actions.
- **Results on real hardware:** across hard drives, mini PCs and laptops (45 physical trials), raised task completion from **58.1% to 86.0%** vs. a scripted baseline and cut cascading failures by **~85%**.
- **Building an NVIDIA Isaac Sim digital twin** of the cell via an isolated URDF→USD export pipeline and ROS 2 bridge, as the basis for simulation-based policy training and sim-to-real evaluation (in progress).
- **Predecessor system (eGRAP):** graph-based adaptive planner for dual-arm HDD disassembly, published at IEEE/ASME AIM 2026.

PUBLICATIONS

[1] **A. R. Das** et al., “Language-Agent Planning over Structured World States for Adaptive Dual-arm Robotic Disassembly,” *IEEE Robotics and Automation Letters*, under review, 2026.

[2] **A. R. Das**, X. Kong and M. Koskinopoulou, “eGRAP: Graph-Based Adaptive Planning for Dual-arm Robotic Disassembly of Electronic Devices,” *IEEE/ASME Int. Conf. on Advanced Intelligent Mechatronics (AIM)*, Genova, 2026.
[doi:10.1109/AIM65483.2026.11658063](https://doi.org/10.1109/AIM65483.2026.11658063)

[3] **A. R. Das** and M. Koskinopoulou, “Toward Sustainable Manufacturing: A Review on Innovations in Robotic Assembly and Disassembly,” *IEEE Access*, vol. 13, pp. 100149–100166, 2025. [doi:10.1109/ACCESS.2025.3576441](https://doi.org/10.1109/ACCESS.2025.3576441)

[4] L. J. Arciniegas-Mayag et al., “Cable-Driven Exosuit to Assist Affected Upper-Limb Users with Hemiparesis,” *IEEE RAS/EMBS Int. Conf. on Biomedical Robotics and Biomechatronics (BioRob)*, Heidelberg, 2024, pp. 1629–1634.
[doi:10.1109/BioRob60516.2024.10719697](https://doi.org/10.1109/BioRob60516.2024.10719697)

PROFESSIONAL EXPERIENCE

- Robotics Engineer — GI Healthcare Industries Ltd, UK**

Dec 2022 – Present

AutoWok semi-autonomous deployable cooking platform · Python, C++, Fusion 360, Tkinter
 - Designed and developed the AutoWok autonomous cooking robot from first prototype, owning both the control back end and the Tkinter operator interface.
 - Ran iterative design research and testing to optimise cooking performance, reliability and operator workflow.
 - Contributed to advancing the platform from **TRL 4 to TRL 6** (2024–2025) under UK Defence and Security Accelerator (DASA) funding; rapid-prototyped mechanisms with Fusion 360 and 3D printing.
- Robotics Intern — Fortrax Robotics, Hyderabad, India**

Mar 2022 – Apr 2022

- Implemented ROS navigation (SLAM) and MoveIt end to end on a custom autonomous mobile robot with a 3-DoF manipulator ([demo](#)).
- Robotics Intern — Rigbetel Labs, Pune, India**

Dec 2021 – Jan 2022

- Independently built a simulated autonomous waiter robot: Fusion 360 design, ROS navigation stack, Gazebo environment and PyQt5 interface ([video](#)).
- Software Engineer — Accenture Solutions Pvt. Ltd., Bangalore, India**

Aug 2020 – Jun 2022

- Developed Selenium/Cucumber test-automation frameworks in Java; designed and executed automated test scenarios across releases.
 - Independently owned a critical module, taking end-to-end responsibility for its software testing and quality sign-off.
- Machine Learning Intern — HighRadius Corporation, Bhubaneswar, India**

Dec 2019 – May 2020

- Ran comprehensive experiments across diverse machine-learning algorithms in scikit-learn to benchmark model performance.
 - Applied feature extraction and feature creation techniques to improve model effectiveness.

TECHNICAL SKILLS

Robotics Software	ROS 2 (Humble), ROS 1, MoveIt 2, EXOTica, URDF/Xacro, SLAM & Nav stack, force-torque sensing
Robot Learning & AI	PyTorch, imitation learning, Vision-Language-Action (VLA) models, LLM agents (LangGraph, ReAct), scikit-learn
Perception	RF-DETR, YOLOv11, Roboflow dataset pipelines, RGB-D perception (Intel RealSense D455, Orbbec Femto Bolt)
Simulation	NVIDIA Isaac Sim (URDF→USD, ROS 2 bridge), PyBullet, Gazebo
Robot Hardware	UFactory UF850 & xArm5, OnRobot RG6, Robotiq FT300, custom end-effectors, 3D-printed fixtures
Programming	Python, C++, Java; Tkinter, PyQt5, HTML/CSS; Git, Linux
CAD & Fabrication	Fusion 360, Onshape, Blender, FDM 3D printing
Languages	English (C2), Hindi (Proficient)

AWARDS & CERTIFICATIONS

- Bristol PLUS Award** — University of Bristol, 2023
- ROS 2 Developer** — Rigbetel Labs LLP, 2023; **ROS 1 Developer** — Rigbetel Labs LLP, 2021
- Fusion 360: Design for Mechatronics** — LinkedIn Learning, 2022